

Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 7: Plant Hormones: What Controls Growth in Maize and Bean Seedlings?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	C	D	D	C	D	A

USE AFTER: OBSERVE PHASE — Observe: hormone evidence table (video and reading on plant hormones)

Q1. A farmer in Kakamega plants bean seeds on the same day after rain. One healthy-looking seed has not germinated because it is still dormant. Which hormone could help break its dormancy and start germination?

- A. Absciscic acid
- B. Ethylene
- C. Gibberellins**
- D. Auxin from the stem tip only

Correct answer: C. Gibberellins break dormancy and promote germination. Absciscic acid maintains dormancy, so it would not start germination.

USE AFTER: OBSERVE PHASE — Observe: hormone evidence table, row for absciscic acid

Q2. Which hormone keeps a seed dormant and also causes the stomata of a leaf to close?

- A. Ethylene
- B. Cytokinin
- C. Absciscic acid**
- D. Gibberellin

Correct answer: C. Absciscic acid is responsible for dormancy and stomatal closure. Gibberellins do the opposite for dormancy, and cytokinins act in cell division.

USE AFTER: OBSERVE PHASE — Observe: highlighting an example for each hormone (ethylene: bananas or mangoes sold in Mombasa)

Q3. Green bananas bought by a trader in Mombasa turn yellow and soft as they ripen. Which plant hormone controls this ripening?

- A. Cytokinin
- B. Gibberellin
- C. Auxin
- D. Ethylene**

Correct answer: D. Ethylene is the hormone that stimulates fruit ripening, for example in bananas and mangoes.

USE AFTER: EXPLAIN PHASE — Explain: CER for (C) bean plants that develop side branches while maize stays unbranched, and the teacher's modelled CER on the tip of the bean stem

Q4. A farmer cuts off the tip of a bean stem and soon side shoots start to grow from the lateral buds. What best explains this?

- A. The tip makes absciscic acid that feeds the side buds, so removing it stops branching
- B. The tip absorbs the rainwater, so removing it leaves more water for the side buds
- C. Cutting the tip makes the plant stop making any hormones
- D. The tip makes auxin that suppresses lateral buds, so removing it lets side buds grow**

Correct answer: D. This is apical dominance: auxin from the tip suppresses lateral buds. When the tip is removed, auxin falls and lateral buds grow, helped by cytokinins.

USE AFTER: EXPLAIN PHASE — Explain: CER rubric oral probes on misconceptions (hormones are food for the plant)

Q5. A student writes: 'Hormones are food for the plant, so a plant with more hormones has more energy to grow.' What is wrong with this idea?

- A. Hormones only work on seeds and never on shoots or roots
- B. Nothing is wrong; hormones are the main food of a plant
- C. Hormones are chemical messengers that control growth processes; they are not food**
- D. Hormones are found only in animals, not in plants

Correct answer: C. Plant hormones are chemical messengers that coordinate processes such as cell division, elongation and germination. Plants make their own food by photosynthesis.

USE AFTER: EXPLAIN PHASE — Explain: CER for (D) farmers in Kericho or Nyeri rooting tea or tree cuttings with natural hormone sources

Q6. Farmers in Kericho root tea cuttings using natural sources such as bulb onion or Aloe extract. Which hormone group do these sources help to supply so that roots form on the cuttings?

- A. Absciscic acid, which keeps the cutting dormant
- B. Ethylene, which ripens the cutting
- C. Gibberellins, which close the stomata
- D. Auxins, which promote the rooting of cuttings**

Correct answer: D. Auxins promote root formation on cuttings, which is why natural rooting sources are used in tree nurseries.

USE AFTER: MODEL BUILDING PHASE — Model: revising the Day 1 model to add hormones and links to dormancy and germination conditions

Q7. Two maize seeds are planted on the same day with the same rain. One germinates in three days and the other takes longer. What is the best explanation that includes hormones?

- A. Hormones respond to conditions such as water, oxygen and temperature, so seeds can differ in when they germinate**
- B. Only seeds without hormones can germinate late
- C. Rain alone decides when every seed germinates, so hormones play no part
- D. Hormones replace water, oxygen and temperature, so these factors do not matter

Correct answer: A. Hormones do not replace environmental conditions; they respond to them. Differences in hormone balance, such as gibberellins versus abscisic acid, can change germination time even with the same rain.